

# GABRIEL CARATIHAN

## SOFTWARE ENGINEER & BUSINESS AUTOMATION SPECIALIST

Communication & Hiring: Available Exclusively via Upwork | Location: Imus, Cavite, PH

### EXECUTIVE SUMMARY

Software Engineer passionate about replacing messy spreadsheets and manual processes with custom, automated web portals. Former ESL Educator turned Systems Architect with a dual perspective that bridges non-technical business leaders seeking operational ROI with robust, enterprise-grade cloud software architecture.

### CORE TECHNICAL SKILLS & BUSINESS FOCUS

#### Languages & Frontend

- TypeScript / JavaScript
- React / Next.js (App Router)
- Tailwind CSS v4 / HTML5
- Framer Motion UI

#### Backend & AI Infrastructure

- Supabase / pgvector
- Google Gemini API (RAG)
- Node.js / REST APIs
- Firebase / Cloud Firestore

#### Business Automation

- 80%+ Operational Time Saved
- Paper-to-Digital Workflows
- RLS & Bank-Grade Security
- Agentic Prompt Engineering

#### &i AGENTIC ENGINEERING & BUSINESS TRANSLATION WORKFLOW

Architected applications using advanced agentic workflows via the Antigravity IDE. Directed AI models (Gemini 3.5 Flash / 3.1 Pro) to rapidly prototype UI/UX, establish PostgreSQL & NoSQL data schemas, enforce Row Level Security (RLS) policies, and automate end-to-end business logic, reducing development lifecycles significantly.

### EDUCATION & PROFESSIONAL BACKGROUND

#### Computer Programming Practitioner

Imus Computer College — Systems & Software Development

#### ESL Educator

Washington School — Educational Technology & Workflows

# GABRIEL CARATIHAN

## SOFTWARE ENGINEER & BUSINESS AUTOMATION SPECIALIST

Communication & Hiring: Available Exclusively via Upwork | Location: Imus, Cavite, PH

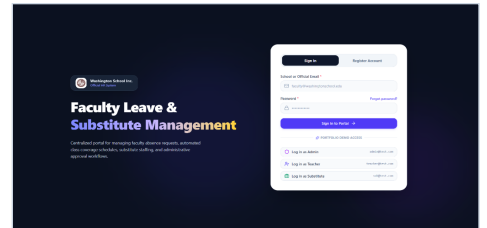
### FEATURED PROJECTS & BUSINESS ROI IMPACT

#### 1. Mini AI DocuMind – Smart Knowledge Base (NEW & FEATURED)

*ROI: 95% Faster Information Retrieval & Zero AI Hallucinations*

Category: AI & RAG Vector Engine | Live Demo & Repository Code Available via Upwork

- Built RAG Q&A assistant for SMEs converting company SOP handbooks into searchable vector embeddings.
- Integrated Google Gemini gemini-embedding-2 with Supabase pgvector cosine similarity search.
- Restricted LLM answers strictly to verified company documents, returning clickable source citations.

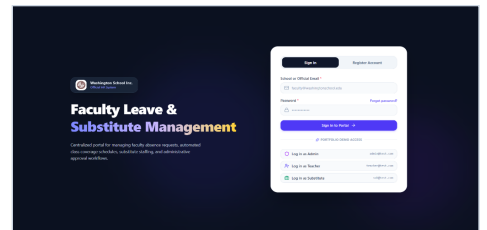


#### 2. Faculty Leave & Substitute Portal (FEATURED)

*ROI: 90% Faster Absence Processing & 100% Automated Sub Coverage*

Category: Enterprise HR System | Live Demo & Repository Code Available via Upwork

- Replaced manual paper leave forms and phone calls with self-service substitute claiming.
- Engineered PostgreSQL RLS policies & DB triggers to automate leave balance deductions.
- Secured lesson plans with 60-second cryptographic URLs and real-time Postgres WebSocket sync.

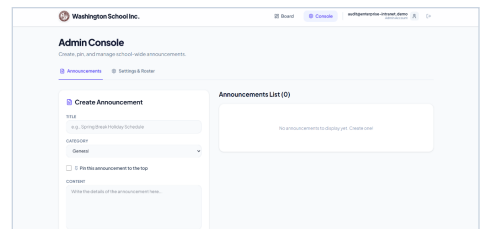


#### 3. Enterprise Communications Intranet

*ROI: 100% Policy Compliance Verification & Zero Lost Announcements*

Category: Compliance & Intranet Portal | Live Demo & Repository Code Available via Upwork

- Replaced unorganized email blasts with digital read-receipt tracking & instant CSV compliance logs.
- Implemented Role-Based Access Control (RBAC) with Cloud Firestore Security Rules.
- Streamlined institutional announcement workflows with secure multi-role administrative oversight.

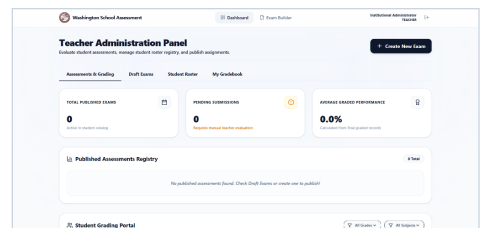


#### 4. Washington Assessment Portal

*ROI: Saved Teachers 15+ Hours/Week in Manual Grading*

Category: EdTech & Assessment LMS | Live Demo & Repository Code Available via Upwork

- Architected dynamic exam creation engine utilizing polymorphic React form components.
- Engineered real-time pivot-table data aggregation for Master Gradebooks & administrative analytics.
- Enforced strict multi-tenant data isolation and time-gated client routing for exam security.



# GABRIEL CARATIHAN

## SOFTWARE ENGINEER & BUSINESS AUTOMATION SPECIALIST

Communication & Hiring: Available Exclusively via Upwork | Location: Imus, Cavite, PH

### 5. AttendancePro Tracker

*ROI: 80% Reduction in Daily Attendance Logging Time*

**Category: Attendance & Early Warning Engine | Live Demo & Repository Code Available via Upwork**

- Replaced physical attendance rosters with proactive truancy algorithms that flag at-risk students.
- Optimized NoSQL database reads using session-grouped data modeling for sub-second roster logging.
- Engineered automated instruction-loss calculations to assist school administrators with intervention.

